

Final Replication Project

Your final assignment for this class will be to replicate and extend an analysis in a recently published academic paper. If you find policy analysis or think tank reports that are rigorous and draw on publicly available data, feel free to use those as well.

1 Due Date and Extension Policy

The assignment must be submitted by **23:59 on Tuesday, May 27**. No extensions are possible, so please give yourself ample time to complete this project.

2 Structure of the Assignment

Consider the format of the labs a template for you to start with. Please submit an .Rmd file (or equivalent with stata or python). Also submit a successfully compiled `html` or `pdf` file.

There are three components to the replication project:

1. Provide a brief summary of the paper, including the elements we stressed in the last quiz:
 - What is the research question?
 - What is the theoretical and empirical estimand
 - What is the dependent variable? What is the treatment?
 - Why is there a causal identification challenge here, and what are the exact sources of confounding?
 - How do the authors propose to address this identification challenge? What method do they use?
 - Illustrate both the challenges and the solution using a DAG and/or potential outcomes notation.
 - What are the assumptions under which their empirical estimand corresponds to the theoretical estimand?
 - What are the assumptions under which the estimate corresponds to the empirical estimand?
 - What are some potential challenges to these assumptions?
2. Replication
 - Tell us which table or figure in the paper you have chosen to replicate. You only need to replicate one, but make sure it's something central to the analysis.¹
 - You must include all code you ran to replicate the table or figure in the provided place. You cannot receive full credit for this part if we cannot see and run all your code.

¹Sometimes papers will first provide descriptive information about their data before getting to the heart of the analysis, which is good practice. Ideally, we're not looking for you to replicate a *descriptive* table or figure, but rather an *analytical* one that speaks to the central causal question. Descriptives are usually averages, analytical tables/figures usually present regression or other modeling output. If in doubt, double-check with us.

- You don't have to write original code here! Ideally, copy-pasting the authors' code will do the trick. We're not looking for original work in this part.
- However, it's possible that a few modifications will be necessary to get the code to run properly. Feel free to make any necessary modifications. You don't have to tell us what modifications you made.
- *Make all necessary modifications using code, not data.* Indeed, you will not submit any data. I expect your code to run and reproduce your results with a freshly downloaded copy of the replication dataset.
- *What if you can't get the numbers to replicate exactly?* You can still get full credit. Explain what you tried and your best hypothesis as to what might be the matter. Maybe you just found a mistake in a published paper!
 - *Don't fabricate numbers!* The task is to *intelligently* replicate the paper using the methods the authors claim they used, not to get the numbers to line up at any cost. You *can* get full credit if the numbers *don't* line up but you made an honest and intelligent effort. You *cannot* get full credit if the numbers *do* line up through lazy or dishonest manipulation.

3. Extension

- This is the fun part. Propose (and execute) an *extension* to the original analysis.
- An extension is something the authors didn't think of, but that would help you assess whether their findings are solid.
- In thinking of an extension, think about potential challenges to causal identification that can be assessed using the provided data. You can usually find datasets linked at the bottom of published articles. If not, a good place to search is the Harvard dataverse.
- We are not looking for anything specific here, and a good extension will depend on what method your paper is using. We recommend looking back at what you learned for that particular method — assumptions, modeling strategies, decisions the researcher has to make, common pitfalls.
- Any extension has to be something you *actually execute with the provided data and your own code*, not just an idea.
- Please justify the extension in the language of causal inference. Explain *why* this might matter for the findings.
- Please briefly conclude with an assessment of what you learned from the paper itself and your own extension. Do the findings hold up to scrutiny?